

EA-0033 - AQELYN Engineering Archive

EA-0033 - AQELYN Identity Security Posture Management (ISPM) Intelligence Engine

****Implementation Specification:**** IS-033

****Publication Standard:**** FULL_COMPLETE

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Engineering Status

Section	Status
Part 1 - Vision, Scope, Requirements	Complete
Part 2 - Components, Integrations, APIs	Complete
Part 3 - Security Architecture, Lifecycles, Performance, Testing, Acceptance Criteria	Complete
Engineering Review	Complete
EA-0033 Package Generation	Complete

Repository Structure Constraint

The AQELYN repository structure is fixed and shall not be redesigned by this Engineering Archive. All artifacts generated by this archive are publication artifacts and do not alter the canonical repository layout.

```
AQELYN/  
■■■■ archive/  
■■■■ blueprint/  
■■■■ docs/  
■■■■ src/  
■■■■ tests/  
■■■■ tools/  
■■■■ build/  
■■■■ releases/  
■■■■ scripts/  
■■■■ assets/  
■■■■ examples/  
■■■■ plugins/  
■■■■ sdk/  
■■■■ api/  
■■■■ README.md
```

1. Executive Summary

The AQELYN Identity Security Posture Management Intelligence Engine provides the identity posture intelligence layer for the AQELYN Cyber Security Operating Environment. It continuously discovers identities, normalizes source data, evaluates posture, calculates risk, detects drift, maps findings to compliance obligations, and recommends remediation. The engine does not act as an identity provider and does not directly mutate external identity systems. Instead, it produces trusted posture intelligence, evidence, events and remediation instructions for downstream AQELYN services.

The purpose of EA-0033 is to provide enough engineering detail for implementation teams to build the engine consistently after the broader AQELYN architecture is complete. The package therefore includes normative requirements, architecture components, data models, event contracts, API contracts, security controls, lifecycle rules, performance design, validation strategy, traceability, examples and diagrams.

The archive is designed to be implementation-ready. A team can derive backlog items, source modules, unit tests, integration tests and acceptance tests directly from this package.

2. Scope

2.1 In Scope

- Identity discovery from enterprise, cloud, workload and platform identity sources.
- Normalization into AQELYN Universal Object Model identity records.
- Identity classification, ownership and lifecycle state tracking.
- Authentication, authorization, monitoring, governance and lifecycle posture checks.
- Identity posture scoring and risk categorization.
- Drift detection against approved identity baselines.
- Compliance and policy mapping.
- Evidence association for every material posture finding.
- Event publication to AQELYN Event Bus.
- Knowledge Graph relationship updates.
- Remediation recommendation generation.
- Historical analytics for trend, delta and effectiveness reporting.

2.2 Out of Scope

- Direct modification of identity provider configuration.
- Replacement of existing IAM, IGA, PAM, CIEM or SIEM products.
- User authentication flows for AQELYN itself.
- Secret vault implementation.
- Manual governance approval workflows, except as integration targets.

3. Engineering Principles

1. Master Markdown is the source of truth.
2. PDF and HTML are generated from the same Markdown.
3. Every requirement must map to architecture, implementation module, test and evidence.
4. The engine must be deterministic for identical input, rule version and baseline version.
5. Connectors are isolated from core scoring logic.
6. Findings must be evidence-backed.
7. Events are immutable facts, not mutable state.
8. Identity posture must be explainable.
9. Risk prioritization must incorporate mission, trust, asset and compliance context.

10. Repository structure remains immutable.

4. Requirements Matrix

ID	Name	Requirement	Component	Verification
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REQ-FR-033-001	Identity discovery	Discover identities from supported identity providers through connector integrations.	ARC-033-001	TC-033-001
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REQ-FR-033-002	Continuous synchronization	Support scheduled, incremental and event-driven synchronization.	ARC-033-002	TC-033-002
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REQ-FR-033-003	Identity normalization	Transform provider records into AQELYN Universal Object Model objects.	ARC-033-003	TC-033-003
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REQ-FR-033-004	Identity inventory	Maintain normalized inventory of all discovered identities with provenance.	ARC-033-004	TC-033-004
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REQ-FR-033-005	Identity classification	Classify identities as human, service, machine, application, federated or temporary.	ARC-033-005	TC-033-005
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REQ-FR-033-006	Authentication assessment	Evaluate MFA, passwordless, FIDO2, credential age and legacy authentication.	ARC-033-006	TC-033-006
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REQ-FR-033-007	Authorization assessment	Evaluate roles, privilege grants, toxic combinations and excess access.	ARC-033-006	TC-033-007
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REQ-FR-033-008	Lifecycle assessment	Detect dormant, orphaned, expired, disabled and ownerless identities.	ARC-033-006	TC-033-008
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REQ-FR-033-009	Policy evaluation	Evaluate identities against policy controls from the AQELYN Policy Engine.	ARC-033-012	TC-033-009
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REQ-FR-033-010	Compliance evaluation	Map posture findings to compliance obligations and control evidence.	ARC-033-012	TC-033-010
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REQ-FR-033-011	Risk scoring	Calculate deterministic Identity Security Posture Score from 0 to 100.	ARC-033-007	TC-033-011
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REQ-FR-033-012	Drift detection	Detect deviations between current state and approved identity baselines.	ARC-033-008	TC-033-012
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REQ-FR-033-013	Recommendation generation	Generate prioritized remediation recommendations for detected findings.	ARC-033-009	TC-033-013
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REQ-FR-033-014	Event publication	Publish identity posture events to the AQELYN Event Bus.	ARC-033-014	TC-033-014
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REQ-FR-033-015	Historical tracking	Maintain historical posture assessments and queryable trends.	ARC-033-015	TC-033-015
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REQ-FR-033-016	Evidence association	Associate every finding with evidence stored by AQELYN Evidence Engine.	ARC-033-010	TC-033-016
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REQ-FR-033-017	Knowledge graph integration	Update identity relationships in the AQELYN Knowledge Graph.	ARC-033-011	TC-033-017
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REQ-FR-033-018	Trust integration	Consume Trust Scores as weighted inputs for posture evaluation.	ARC-033-013	TC-033-018
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REQ-FR-033-019	Workflow integration	Initiate remediation workflows when configured.	ARC-033-009	TC-033-019
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REQ-FR-033-020	Mission awareness	Prioritize posture risk using mission criticality.	ARC-033-013	TC-033-020
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ID	Name	Requirement	Verification	Evidence
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REQ-NFR-033-001	Availability	Maintain 99.9% service availability through redundant workers and graceful degradation.	Availability test	EV-033-021
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REQ-NFR-033-002	Scalability	Support at least 10 million identities through partitioned inventory and horizontal workers.	Load test	EV-033-022
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REQ-NFR-033-003	Latency	Assess one identity in <=5 seconds under nominal load.	Performance test	EV-033-023
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REQ-NFR-033-004	Extensibility	Add providers through plug-in connectors without modifying core assessment logic.	Connector test	EV-033-024
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REQ-NFR-033-005	Determinism	Produce identical scores for identical input, rule and baseline versions.	Repeatability test	EV-033-025
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REQ-NFR-033-006	Auditability	Persist immutable assessment, drift and recommendation history.	Audit review	EV-033-026
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REQ-NFR-033-007	Security	Protect identity metadata using least privilege, encryption and secret isolation.	Security test	EV-033-027
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REQ-NFR-033-008	Observability	Emit structured logs, metrics and traces for all processing stages.	Operational test	EV-033-028
REQ-NFR-033-009	Resilience	Connector failure shall not halt unrelated assessments.	Fault injection	EV-033-029
REQ-NFR-033-010	Maintainability	Version all rule sets, schemas and connector contracts.	Code review	EV-033-030

5. Logical Architecture

The engine consists of provider-facing ingestion components, canonical identity state services, analytic engines, integration adapters and publication services. Components communicate through internal contracts and emit events through the AQELYN Event Bus.

| Component ID | Component | Responsibility |

| --- | --- | --- |

| ARC-033-001 | Identity Discovery Manager | Discovers human, service, machine, application, federated and temporary identities from provider connectors. |

| ARC-033-002 | Connector Orchestration Layer | Schedules connector jobs, isolates failures, handles cursor state and normalizes connector telemetry. |

| ARC-033-003 | Identity Normalization Service | Transforms provider-specific identity records into AQELYN Universal Object Model instances. |

| ARC-033-004 | Identity Inventory Repository | Stores current identity state, ownership, provenance, lifecycle data and posture state. |

| ARC-033-005 | Identity Classification Engine | Classifies identities and applies confidence scoring and conflict resolution. |

| ARC-033-006 | Posture Assessment Engine | Evaluates authentication, authorization, lifecycle, governance and monitoring controls. |

| ARC-033-007 | Risk Scoring Engine | Calculates deterministic 0-100 posture scores and normalized risk ratings. |

| ARC-033-008 | Drift Detection Engine | Compares current state to approved baselines and emits drift findings. |

| ARC-033-009 | Recommendation Engine | Generates prioritized remediation guidance with evidence and workflow hints. |

| ARC-033-010 | Evidence Integration Layer | Links findings, scores and recommendations to immutable evidence records. |

| ARC-033-011 | Knowledge Graph Publisher | Publishes identity relationships, ownership edges, privilege edges and dependency edges. |

| ARC-033-012 | Policy and Compliance Adapter | Consumes policy/compliance rules and maps identity findings to control obligations. |

| ARC-033-013 | Trust and Mission Context Adapter | Consumes trust scores and mission criticality for risk prioritization. |

| ARC-033-014 | Event Publication Layer | Publishes canonical identity posture events to the AQELYN Event Bus. |

| ARC-033-015 | Historical Analytics Store | Maintains posture trends, score deltas, remediation effectiveness and drift history. |

6.1 Component Specification - Identity Discovery Manager

Component ID: ARC-033-001

Purpose

Discovers human, service, machine, application, federated and temporary identities from provider connectors.

Responsibilities

- Accept only versioned input contracts.
- Validate all required fields and reject malformed records with structured errors.
- Emit operational telemetry for success, failure, latency and retries.
- Preserve source provenance and correlation identifiers.
- Avoid direct coupling to external components except through declared interfaces.

Inputs

- Canonical AQELYN events.
- Provider-specific payloads when the component is a connector boundary.
- Policy, trust, compliance, mission and evidence references where applicable.

Outputs

- Canonical identity posture objects.
- Immutable findings, scores, drift records and recommendations.
- Evidence references and event bus messages.

Failure Handling

Failures are classified as validation failures, provider failures, transient infrastructure failures, dependency failures or internal logic failures. Transient failures use bounded retry. Provider failures are isolated by connector and never halt unrelated providers.

Implementation Notes

The implementation module corresponding to ARC-033-001 shall expose a stable interface, unit tests, integration tests and structured logs. Module boundaries must match the traceability matrix so that engineering evidence remains auditable.

6.2 Component Specification - Connector Orchestration Layer

Component ID: ARC-033-002

Purpose

Schedules connector jobs, isolates failures, handles cursor state and normalizes connector telemetry.

Responsibilities

- Accept only versioned input contracts.
- Validate all required fields and reject malformed records with structured errors.
- Emit operational telemetry for success, failure, latency and retries.
- Preserve source provenance and correlation identifiers.
- Avoid direct coupling to external components except through declared interfaces.

Inputs

- Canonical AQELYN events.
- Provider-specific payloads when the component is a connector boundary.
- Policy, trust, compliance, mission and evidence references where applicable.

Outputs

- Canonical identity posture objects.
- Immutable findings, scores, drift records and recommendations.

- Evidence references and event bus messages.

Failure Handling

Failures are classified as validation failures, provider failures, transient infrastructure failures, dependency failures or internal logic failures. Transient failures use bounded retry. Provider failures are isolated by connector and never halt unrelated providers.

Implementation Notes

The implementation module corresponding to ARC-033-002 shall expose a stable interface, unit tests, integration tests and structured logs. Module boundaries must match the traceability matrix so that engineering evidence remains auditable.

6.3 Component Specification - Identity Normalization Service

****Component ID:**** ARC-033-003

Purpose

Transforms provider-specific identity records into AQELYN Universal Object Model instances.

Responsibilities

- Accept only versioned input contracts.
- Validate all required fields and reject malformed records with structured errors.
- Emit operational telemetry for success, failure, latency and retries.
- Preserve source provenance and correlation identifiers.
- Avoid direct coupling to external components except through declared interfaces.

Inputs

- Canonical AQELYN events.
- Provider-specific payloads when the component is a connector boundary.
- Policy, trust, compliance, mission and evidence references where applicable.

Outputs

- Canonical identity posture objects.
- Immutable findings, scores, drift records and recommendations.
- Evidence references and event bus messages.

Failure Handling

Failures are classified as validation failures, provider failures, transient infrastructure failures, dependency failures or internal logic failures. Transient failures use bounded retry. Provider failures are isolated by connector and never halt unrelated providers.

Implementation Notes

The implementation module corresponding to ARC-033-003 shall expose a stable interface, unit tests, integration tests and structured logs. Module boundaries must match the traceability matrix so that engineering evidence remains auditable.

6.4 Component Specification - Identity Inventory Repository

****Component ID:**** ARC-033-004

Purpose

Stores current identity state, ownership, provenance, lifecycle data and posture state.

Responsibilities

- Accept only versioned input contracts.
- Validate all required fields and reject malformed records with structured errors.
- Emit operational telemetry for success, failure, latency and retries.
- Preserve source provenance and correlation identifiers.
- Avoid direct coupling to external components except through declared interfaces.

Inputs

- Canonical AQELYN events.
- Provider-specific payloads when the component is a connector boundary.
- Policy, trust, compliance, mission and evidence references where applicable.

Outputs

- Canonical identity posture objects.
- Immutable findings, scores, drift records and recommendations.
- Evidence references and event bus messages.

Failure Handling

Failures are classified as validation failures, provider failures, transient infrastructure failures, dependency failures or internal logic failures. Transient failures use bounded retry. Provider failures are isolated by connector and never halt unrelated providers.

Implementation Notes

The implementation module corresponding to ARC-033-004 shall expose a stable interface, unit tests, integration tests and structured logs. Module boundaries must match the traceability matrix so that engineering evidence remains auditable.

6.5 Component Specification - Identity Classification Engine

****Component ID:**** ARC-033-005

Purpose

Classifies identities and applies confidence scoring and conflict resolution.

Responsibilities

- Accept only versioned input contracts.
- Validate all required fields and reject malformed records with structured errors.
- Emit operational telemetry for success, failure, latency and retries.
- Preserve source provenance and correlation identifiers.
- Avoid direct coupling to external components except through declared interfaces.

Inputs

- Canonical AQELYN events.
- Provider-specific payloads when the component is a connector boundary.
- Policy, trust, compliance, mission and evidence references where applicable.

Outputs

- Canonical identity posture objects.
- Immutable findings, scores, drift records and recommendations.
- Evidence references and event bus messages.

Failure Handling

Failures are classified as validation failures, provider failures, transient infrastructure failures, dependency failures or internal logic failures. Transient failures use bounded retry. Provider failures are isolated by connector and never halt unrelated providers.

Implementation Notes

The implementation module corresponding to ARC-033-005 shall expose a stable interface, unit tests, integration tests and structured logs. Module boundaries must match the traceability matrix so that engineering evidence remains auditable.

6.6 Component Specification - Posture Assessment Engine

****Component ID:**** ARC-033-006

Purpose

Evaluates authentication, authorization, lifecycle, governance and monitoring controls.

Responsibilities

- Accept only versioned input contracts.
- Validate all required fields and reject malformed records with structured errors.
- Emit operational telemetry for success, failure, latency and retries.
- Preserve source provenance and correlation identifiers.

- Avoid direct coupling to external components except through declared interfaces.

Inputs

- Canonical AQELYN events.
- Provider-specific payloads when the component is a connector boundary.
- Policy, trust, compliance, mission and evidence references where applicable.

Outputs

- Canonical identity posture objects.
- Immutable findings, scores, drift records and recommendations.
- Evidence references and event bus messages.

Failure Handling

Failures are classified as validation failures, provider failures, transient infrastructure failures, dependency failures or internal logic failures. Transient failures use bounded retry. Provider failures are isolated by connector and never halt unrelated providers.

Implementation Notes

The implementation module corresponding to ARC-033-006 shall expose a stable interface, unit tests, integration tests and structured logs. Module boundaries must match the traceability matrix so that engineering evidence remains auditable.

6.7 Component Specification - Risk Scoring Engine

****Component ID:**** ARC-033-007

Purpose

Calculates deterministic 0-100 posture scores and normalized risk ratings.

Responsibilities

- Accept only versioned input contracts.
- Validate all required fields and reject malformed records with structured errors.
- Emit operational telemetry for success, failure, latency and retries.
- Preserve source provenance and correlation identifiers.
- Avoid direct coupling to external components except through declared interfaces.

Inputs

- Canonical AQELYN events.
- Provider-specific payloads when the component is a connector boundary.
- Policy, trust, compliance, mission and evidence references where applicable.

Outputs

- Canonical identity posture objects.
- Immutable findings, scores, drift records and recommendations.
- Evidence references and event bus messages.

Failure Handling

Failures are classified as validation failures, provider failures, transient infrastructure failures, dependency failures or internal logic failures. Transient failures use bounded retry. Provider failures are isolated by connector and never halt unrelated providers.

Implementation Notes

The implementation module corresponding to ARC-033-007 shall expose a stable interface, unit tests, integration tests and structured logs. Module boundaries must match the traceability matrix so that engineering evidence remains auditable.

6.8 Component Specification - Drift Detection Engine

****Component ID:**** ARC-033-008

Purpose

Compares current state to approved baselines and emits drift findings.

Responsibilities

- Accept only versioned input contracts.
- Validate all required fields and reject malformed records with structured errors.
- Emit operational telemetry for success, failure, latency and retries.
- Preserve source provenance and correlation identifiers.
- Avoid direct coupling to external components except through declared interfaces.

Inputs

- Canonical AQELYN events.
- Provider-specific payloads when the component is a connector boundary.
- Policy, trust, compliance, mission and evidence references where applicable.

Outputs

- Canonical identity posture objects.
- Immutable findings, scores, drift records and recommendations.
- Evidence references and event bus messages.

Failure Handling

Failures are classified as validation failures, provider failures, transient infrastructure failures, dependency failures or internal logic failures. Transient failures use bounded retry. Provider failures are isolated by connector and never halt

unrelated providers.

Implementation Notes

The implementation module corresponding to ARC-033-008 shall expose a stable interface, unit tests, integration tests and structured logs. Module boundaries must match the traceability matrix so that engineering evidence remains auditable.

6.9 Component Specification - Recommendation Engine

****Component ID:**** ARC-033-009

Purpose

Generates prioritized remediation guidance with evidence and workflow hints.

Responsibilities

- Accept only versioned input contracts.
- Validate all required fields and reject malformed records with structured errors.
- Emit operational telemetry for success, failure, latency and retries.
- Preserve source provenance and correlation identifiers.
- Avoid direct coupling to external components except through declared interfaces.

Inputs

- Canonical AQELYN events.
- Provider-specific payloads when the component is a connector boundary.
- Policy, trust, compliance, mission and evidence references where applicable.

Outputs

- Canonical identity posture objects.
- Immutable findings, scores, drift records and recommendations.
- Evidence references and event bus messages.

Failure Handling

Failures are classified as validation failures, provider failures, transient infrastructure failures, dependency failures or internal logic failures. Transient failures use bounded retry. Provider failures are isolated by connector and never halt unrelated providers.

Implementation Notes

The implementation module corresponding to ARC-033-009 shall expose a stable interface, unit tests, integration tests and structured logs. Module boundaries must match the traceability matrix so that engineering evidence remains auditable.

6.10 Component Specification - Evidence Integration Layer

****Component ID:**** ARC-033-010

Purpose

Links findings, scores and recommendations to immutable evidence records.

Responsibilities

- Accept only versioned input contracts.
- Validate all required fields and reject malformed records with structured errors.
- Emit operational telemetry for success, failure, latency and retries.
- Preserve source provenance and correlation identifiers.
- Avoid direct coupling to external components except through declared interfaces.

Inputs

- Canonical AQELYN events.
- Provider-specific payloads when the component is a connector boundary.
- Policy, trust, compliance, mission and evidence references where applicable.

Outputs

- Canonical identity posture objects.
- Immutable findings, scores, drift records and recommendations.
- Evidence references and event bus messages.

Failure Handling

Failures are classified as validation failures, provider failures, transient infrastructure failures, dependency failures or internal logic failures. Transient failures use bounded retry. Provider failures are isolated by connector and never halt unrelated providers.

Implementation Notes

The implementation module corresponding to ARC-033-010 shall expose a stable interface, unit tests, integration tests and structured logs. Module boundaries must match the traceability matrix so that engineering evidence remains auditable.

6.11 Component Specification - Knowledge Graph Publisher

****Component ID:**** ARC-033-011

Purpose

Publishes identity relationships, ownership edges, privilege edges and dependency edges.

Responsibilities

- Accept only versioned input contracts.

- Validate all required fields and reject malformed records with structured errors.
- Emit operational telemetry for success, failure, latency and retries.
- Preserve source provenance and correlation identifiers.
- Avoid direct coupling to external components except through declared interfaces.

Inputs

- Canonical AQELYN events.
- Provider-specific payloads when the component is a connector boundary.
- Policy, trust, compliance, mission and evidence references where applicable.

Outputs

- Canonical identity posture objects.
- Immutable findings, scores, drift records and recommendations.
- Evidence references and event bus messages.

Failure Handling

Failures are classified as validation failures, provider failures, transient infrastructure failures, dependency failures or internal logic failures. Transient failures use bounded retry. Provider failures are isolated by connector and never halt unrelated providers.

Implementation Notes

The implementation module corresponding to ARC-033-011 shall expose a stable interface, unit tests, integration tests and structured logs. Module boundaries must match the traceability matrix so that engineering evidence remains auditable.

6.12 Component Specification - Policy and Compliance Adapter

Component ID: ARC-033-012

Purpose

Consumes policy/compliance rules and maps identity findings to control obligations.

Responsibilities

- Accept only versioned input contracts.
- Validate all required fields and reject malformed records with structured errors.
- Emit operational telemetry for success, failure, latency and retries.
- Preserve source provenance and correlation identifiers.
- Avoid direct coupling to external components except through declared interfaces.

Inputs

- Canonical AQELYN events.

- Provider-specific payloads when the component is a connector boundary.
- Policy, trust, compliance, mission and evidence references where applicable.

Outputs

- Canonical identity posture objects.
- Immutable findings, scores, drift records and recommendations.
- Evidence references and event bus messages.

Failure Handling

Failures are classified as validation failures, provider failures, transient infrastructure failures, dependency failures or internal logic failures. Transient failures use bounded retry. Provider failures are isolated by connector and never halt unrelated providers.

Implementation Notes

The implementation module corresponding to ARC-033-012 shall expose a stable interface, unit tests, integration tests and structured logs. Module boundaries must match the traceability matrix so that engineering evidence remains auditable.

6.13 Component Specification - Trust and Mission Context Adapter

****Component ID:**** ARC-033-013

Purpose

Consumes trust scores and mission criticality for risk prioritization.

Responsibilities

- Accept only versioned input contracts.
- Validate all required fields and reject malformed records with structured errors.
- Emit operational telemetry for success, failure, latency and retries.
- Preserve source provenance and correlation identifiers.
- Avoid direct coupling to external components except through declared interfaces.

Inputs

- Canonical AQELYN events.
- Provider-specific payloads when the component is a connector boundary.
- Policy, trust, compliance, mission and evidence references where applicable.

Outputs

- Canonical identity posture objects.
- Immutable findings, scores, drift records and recommendations.

- Evidence references and event bus messages.

Failure Handling

Failures are classified as validation failures, provider failures, transient infrastructure failures, dependency failures or internal logic failures. Transient failures use bounded retry. Provider failures are isolated by connector and never halt unrelated providers.

Implementation Notes

The implementation module corresponding to ARC-033-013 shall expose a stable interface, unit tests, integration tests and structured logs. Module boundaries must match the traceability matrix so that engineering evidence remains auditable.

6.14 Component Specification - Event Publication Layer

****Component ID:**** ARC-033-014

Purpose

Publishes canonical identity posture events to the AQELYN Event Bus.

Responsibilities

- Accept only versioned input contracts.
- Validate all required fields and reject malformed records with structured errors.
- Emit operational telemetry for success, failure, latency and retries.
- Preserve source provenance and correlation identifiers.
- Avoid direct coupling to external components except through declared interfaces.

Inputs

- Canonical AQELYN events.
- Provider-specific payloads when the component is a connector boundary.
- Policy, trust, compliance, mission and evidence references where applicable.

Outputs

- Canonical identity posture objects.
- Immutable findings, scores, drift records and recommendations.
- Evidence references and event bus messages.

Failure Handling

Failures are classified as validation failures, provider failures, transient infrastructure failures, dependency failures or internal logic failures. Transient failures use bounded retry. Provider failures are isolated by connector and never halt unrelated providers.

Implementation Notes

The implementation module corresponding to ARC-033-014 shall expose a stable interface, unit tests, integration tests and structured logs. Module boundaries must match the traceability matrix so that engineering evidence remains auditable.

6.15 Component Specification - Historical Analytics Store

****Component ID:**** ARC-033-015

Purpose

Maintains posture trends, score deltas, remediation effectiveness and drift history.

Responsibilities

- Accept only versioned input contracts.
- Validate all required fields and reject malformed records with structured errors.
- Emit operational telemetry for success, failure, latency and retries.
- Preserve source provenance and correlation identifiers.
- Avoid direct coupling to external components except through declared interfaces.

Inputs

- Canonical AQELYN events.
- Provider-specific payloads when the component is a connector boundary.
- Policy, trust, compliance, mission and evidence references where applicable.

Outputs

- Canonical identity posture objects.
- Immutable findings, scores, drift records and recommendations.
- Evidence references and event bus messages.

Failure Handling

Failures are classified as validation failures, provider failures, transient infrastructure failures, dependency failures or internal logic failures. Transient failures use bounded retry. Provider failures are isolated by connector and never halt unrelated providers.

Implementation Notes

The implementation module corresponding to ARC-033-015 shall expose a stable interface, unit tests, integration tests and structured logs. Module boundaries must match the traceability matrix so that engineering evidence remains auditable.

7. Canonical Domain Model

The domain model below defines implementation-level objects. Field names are normative for API, event and persistence schemas unless a future architecture decision explicitly supersedes them.

7.1 AQELYNIdentity

```
{
  "identity_id": "uuid",
  "provider":
    "entra_id|active_directory|okta|aws_iam|azure_managed_identity|gcp_iam|kubernetes|ldap|scim|local",
  "provider_identity_id": "string",
  "identity_type": "human|service|machine|application|federated|temporary",
  "display_name": "string",
  "owner_id": "uuid|null",
  "lifecycle_state": "active|inactive|disabled|expired|orphaned|unknown",
  "privilege_level": "none|standard|elevated|privileged|break_glass",
  "mfa_status": "enabled|disabled|not_applicable|unknown",
  "trust_score": 0,
  "posture_score": 0,
  "risk_rating": "excellent|good|moderate|elevated|critical",
  "last_seen_at": "timestamp|null",
  "source_provenance": []
}
```

7.2 IdentityFinding

```
{
  "finding_id": "uuid",
  "identity_id": "uuid",
  "finding_type": "authentication|authorization|lifecycle|governance|monitoring|compliance|behavioral",
  "severity": "info|low|medium|high|critical",
  "title": "string",
  "description": "string",
  "evidence_refs": ["EV-..."],
  "policy_refs": ["POL-..."],
  "recommendation_refs": ["REC-..."],
  "detected_at": "timestamp",
  "rule_version": "semver"
}
```

7.3 IdentityBaseline

```
{
  "baseline_id": "uuid",
  "scope": "tenant|provider|group|identity|mission",
  "approved_by": "string",
  "approved_at": "timestamp",
  "version": "semver",
  "rules": [],
  "expected_state": {}
}
```

8. Scoring Model

The Identity Security Posture Score is a deterministic 0-100 score where 100 means no detected identity posture weakness under the active rule set. Scores are calculated from weighted dimensions.

| Dimension | Weight | Examples |

| --- | --- | --- |

| Authentication Quality | 25% | MFA, phishing-resistant authentication, credential age, legacy auth |

| Authorization Quality | 25% | Admin roles, excessive permissions, toxic combinations |

| Lifecycle Hygiene | 15% | Dormant, orphaned, expired, ownerless accounts |

| Governance and Compliance | 15% | Policy violations, exceptions, approvals |

| Monitoring Coverage | 10% | Audit logs, identity telemetry, detection coverage |

| Behavioral and Trust Signals | 10% | Trust score, anomaly hints, mission context |

8.1 Score Bands

| Score Range | Rating | Required Response |

--- --- ---
90-100 Excellent Normal monitoring
75-89 Good Planned improvement
60-74 Moderate Remediation backlog
40-59 Elevated Priority remediation
0-39 Critical Immediate review and containment planning

8.2 Determinism Rule

Given the same identity state, baseline version, rule version, policy version, trust input and mission context, the scoring engine shall produce the same score and explanation. Time-dependent logic must use the assessment timestamp as an explicit input.

9. Event Model

Events are immutable and published through the AQELYN Event Bus. All events include event_id, event_type, schema_version, occurred_at, producer, correlation_id, causation_id, tenant_id and evidence_refs.

Event Type Purpose Primary Payload
--- --- ---
IdentityDiscovered A new identity was discovered identity_id, provider, provider_identity_id
IdentityInventoryUpdated Identity state changed identity_id, changed_fields
IdentityClassificationChanged Classification changed identity_id, previous_type, new_type
IdentityAssessmentCompleted Assessment completed identity_id, assessment_id, rule_version
IdentityRiskCalculated Score calculated identity_id, score, risk_rating
IdentityDriftDetected Baseline deviation found identity_id, baseline_id, drift_type
IdentityRecommendationGenerated Remediation recommendation created recommendation_id, identity_id, priority
IdentityComplianceViolationDetected Compliance mapping created identity_id, framework, control_id
IdentityPostureScoreChanged Material score delta detected identity_id, previous_score, new_score

10. API Contracts

The public service API is implementation-facing and should be exposed through the AQELYN API layer. Authorization is mandatory for all endpoints.

Method Path Purpose
--- --- ---
GET /api/isp/v1/identities List identities with filters
GET /api/isp/v1/identities/{identity_id} Get canonical identity record
POST /api/isp/v1/assessments Request assessment for scope
GET /api/isp/v1/assessments/{assessment_id} Read assessment results
GET /api/isp/v1/findings Search posture findings
GET /api/isp/v1/recommendations Search remediation recommendations
POST /api/isp/v1/baselines Create or propose baseline
GET /api/isp/v1/drift Search drift records
GET /api/isp/v1/trends Read posture trend analytics

10.1 Error Contract

```
{
  "error_id": "uuid",
  "code": "VALIDATION_ERROR|UNAUTHORIZED|FORBIDDEN|NOT_FOUND|CONFLICT|DEPENDENCY_FAILURE|INTERNAL_ERROR",
  "message": "safe human-readable message",
  "correlation_id": "uuid",
  "details": {}
}
```

11. Security Architecture

The ISPM engine processes sensitive identity metadata. Security controls therefore apply to connectors, storage, APIs, events, logs and generated artifacts.

11.1 Core Controls

- Least-privilege provider credentials scoped to read-only posture assessment wherever possible.
- Secret material stored only in an approved secrets facility, never in logs or events.
- Encryption in transit for all service calls.
- Encryption at rest for inventory, assessment history and evidence references.
- Field-level redaction for sensitive attributes in logs.
- Role-based authorization for API and administrative operations.
- Separation of connector execution roles from scoring and publication roles.
- Immutable audit trail for baselines, rules and assessment outputs.

11.2 Abuse Resistance

The engine must not expose credential material, must not provide instructions for privilege escalation, and must not execute remediation against external identity providers unless a separate approved workflow engine action authorizes the change.

12. Lifecycle Architecture

12.1 Discovery Lifecycle

1. Connector job scheduled or triggered.
2. Provider queried using cursor or full sync.
3. Payload validated and normalized.
4. Inventory updated.
5. Change events emitted.

12.2 Assessment Lifecycle

1. Assessment scope selected.
2. Applicable policies, baselines, rule sets, trust inputs and mission context loaded.
3. Posture checks executed.
4. Score calculated.
5. Findings and evidence references persisted.
6. Events published.

7. Recommendations generated.

12.3 Drift Lifecycle

1. Current identity state retrieved.
2. Approved baseline resolved.
3. Comparator identifies deviation.
4. Drift severity assigned.
5. Drift finding emitted with evidence.
6. Recommendation linked.

13. Performance and Scalability

The implementation must support large identity estates by partitioning inventory by tenant, provider and identity scope. Assessment workers must process independent identity partitions concurrently. Historical analytics should use append-optimized storage and materialized aggregates for dashboards.

Performance Target	Design Response
10 million identities	Partitioned inventory and horizontally scalable workers
<=5 seconds per identity assessment	Precompiled rule evaluation and cached context inputs
Connector failure isolation	Provider-scoped worker pools and retry queues
Large trend queries	Rollup tables and bounded query windows
Event burst handling	Backpressure-aware Event Bus publishing

14. Observability

The engine shall expose metrics, logs and traces for every processing stage.

Signal	Examples
Metrics	connector_success_total, connector_failure_total, assessment_latency_ms, findings_by_severity, score_delta_total
Logs	structured JSON with correlation_id, component_id, provider, tenant_id and safe error code
Traces	discovery trace, normalization trace, assessment trace, publication trace
Dashboards	connector health, posture score distribution, critical identity findings, remediation aging

15. Testing Strategy

Test Suite	Coverage
Unit tests	Normalizers, rule packs, scoring functions, comparators, recommendation mappers
Integration tests	Provider connectors, Event Bus adapter, Evidence Engine, Knowledge Graph, Trust Engine, Policy Engine
System tests	End-to-end discovery to finding to recommendation to event publication
Performance tests	Large inventory, high event rate and scoring throughput
Resilience tests	Provider outage, event bus backpressure, malformed records and dependency failures
Security tests	Authorization, secret handling, log redaction, tenant isolation

| Regression tests | Rule set version compatibility and deterministic scoring |

15.1 Acceptance Criteria

EA-0033 is accepted when each requirement has traceability to an architecture component, implementation module, verification test and evidence artifact, and when the generated package contains master Markdown, complete PDF, complete HTML, README, manifest, matrices, engineering journal, examples, five SVG diagrams and final ZIP.

16. Requirements Traceability Matrix

| Requirement | Architecture Component | Implementation Module | Verification | Evidence |

| --- | --- | --- | --- | --- |

| REQ-FR-033-001 | ARC-033-001 | MOD-033-001 | TC-033-001 | EV-033-001 |

| REQ-FR-033-002 | ARC-033-002 | MOD-033-002 | TC-033-002 | EV-033-002 |

| REQ-FR-033-003 | ARC-033-003 | MOD-033-003 | TC-033-003 | EV-033-003 |

| REQ-FR-033-004 | ARC-033-004 | MOD-033-004 | TC-033-004 | EV-033-004 |

| REQ-FR-033-005 | ARC-033-005 | MOD-033-005 | TC-033-005 | EV-033-005 |

| REQ-FR-033-006 | ARC-033-006 | MOD-033-006 | TC-033-006 | EV-033-006 |

| REQ-FR-033-007 | ARC-033-006 | MOD-033-007 | TC-033-007 | EV-033-007 |

| REQ-FR-033-008 | ARC-033-006 | MOD-033-008 | TC-033-008 | EV-033-008 |

| REQ-FR-033-009 | ARC-033-012 | MOD-033-009 | TC-033-009 | EV-033-009 |

| REQ-FR-033-010 | ARC-033-012 | MOD-033-010 | TC-033-010 | EV-033-010 |

| REQ-FR-033-011 | ARC-033-007 | MOD-033-011 | TC-033-011 | EV-033-011 |

| REQ-FR-033-012 | ARC-033-008 | MOD-033-012 | TC-033-012 | EV-033-012 |

| REQ-FR-033-013 | ARC-033-009 | MOD-033-013 | TC-033-013 | EV-033-013 |

| REQ-FR-033-014 | ARC-033-014 | MOD-033-014 | TC-033-014 | EV-033-014 |

| REQ-FR-033-015 | ARC-033-015 | MOD-033-015 | TC-033-015 | EV-033-015 |

| REQ-FR-033-016 | ARC-033-010 | MOD-033-016 | TC-033-016 | EV-033-016 |

| REQ-FR-033-017 | ARC-033-011 | MOD-033-017 | TC-033-017 | EV-033-017 |

| REQ-FR-033-018 | ARC-033-013 | MOD-033-018 | TC-033-018 | EV-033-018 |

| REQ-FR-033-019 | ARC-033-009 | MOD-033-019 | TC-033-019 | EV-033-019 |

| REQ-FR-033-020 | ARC-033-013 | MOD-033-020 | TC-033-020 | EV-033-020 |

| Requirement | Design Strategy | Verification | Evidence |

| --- | --- | --- | --- |

| REQ-NFR-033-001 | Maintain 99.9% service availability through redundant workers and graceful degradation. | Availability test | EV-033-021 |

| REQ-NFR-033-002 | Support at least 10 million identities through partitioned inventory and horizontal workers. | Load test | EV-033-022 |

| REQ-NFR-033-003 | Assess one identity in <=5 seconds under nominal load. | Performance test | EV-033-023 |

| REQ-NFR-033-004 | Add providers through plug-in connectors without modifying core assessment logic. | Connector test | EV-033-024 |

| REQ-NFR-033-005 | Produce identical scores for identical input, rule and baseline versions. | Repeatability test | EV-033-025 |

| REQ-NFR-033-006 | Persist immutable assessment, drift and recommendation history. | Audit review | EV-033-026 |

| REQ-NFR-033-007 | Protect identity metadata using least privilege, encryption and secret isolation. | Security test | EV-033-027 |

| REQ-NFR-033-008 | Emit structured logs, metrics and traces for all processing stages. | Operational test | EV-033-028 |

| REQ-NFR-033-009 | Connector failure shall not halt unrelated assessments. | Fault injection | EV-033-029 |

| REQ-NFR-033-010 | Version all rule sets, schemas and connector contracts. | Code review | EV-033-030 |

17. Engineering Review

| Review Area | Result | Notes |

| --- | --- | --- |

| Completeness | Pass | All required FULL_COMPLETE sections are present. |

| Repository Compliance | Pass | No repository redesign is introduced. |

| Archive Sequence | Pass | EA-0033 follows EA-0032 and does not skip numbering. |

| Traceability | Pass | Functional and non-functional requirements map to design and verification. |

| Security | Pass | Sensitive identity metadata protections are specified. |

| Implementation Readiness | Pass | Components, contracts, schemas, events and tests are defined. |

18. Implementation Backlog

| Epic | Representative Stories |

| --- | --- |

| Connector Framework | Implement connector SDK, provider adapters, cursor state and health reporting |

| Normalization and Inventory | Implement object mapper, inventory repository, ownership model and provenance tracking |

| Assessment Engine | Implement rule packs for authentication, authorization, lifecycle, governance and monitoring |

| Scoring and Drift | Implement deterministic scoring, baseline comparator and score delta publication |

| Integrations | Implement Event Bus, Evidence, Knowledge Graph, Policy, Compliance, Trust, Mission and Workflow adapters |

| APIs and Analytics | Implement query APIs, trend rollups, dashboards and export contracts |

| Verification | Implement unit, integration, system, performance, resilience and security tests |

19. Example Artifacts

The package includes example JSON and YAML artifacts under the examples directory. These examples are non-production test fixtures that illustrate canonical payloads and rule configuration.

20. Publication Manifest

The manifest.json file in this package contains SHA-256 hashes for all generated files. The manifest is generated after all files are written and should be used to verify archive integrity.

Appendix A.1 Rule Specification - ISPM-RULE-001

Domain: authorization

Default Severity: medium

Intent

This rule detects a material identity posture condition in the authorization domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.

- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.2 Rule Specification - ISPM-RULE-002

Domain: lifecycle

Default Severity: high

Intent

This rule detects a material identity posture condition in the lifecycle domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.3 Rule Specification - ISPM-RULE-003

****Domain:**** governance

****Default Severity:**** critical

Intent

This rule detects a material identity posture condition in the governance domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.4 Rule Specification - ISPM-RULE-004

****Domain:**** monitoring

****Default Severity:**** low

Intent

This rule detects a material identity posture condition in the monitoring domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.5 Rule Specification - ISPM-RULE-005

****Domain:**** compliance

****Default Severity:**** medium

Intent

This rule detects a material identity posture condition in the compliance domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.6 Rule Specification - ISPM-RULE-006

****Domain:**** authentication

****Default Severity:**** high

Intent

This rule detects a material identity posture condition in the authentication domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be

represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.7 Rule Specification - ISPM-RULE-007

Domain: authorization

Default Severity: critical

Intent

This rule detects a material identity posture condition in the authorization domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.

- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.8 Rule Specification - ISPM-RULE-008

****Domain:**** lifecycle

****Default Severity:**** low

Intent

This rule detects a material identity posture condition in the lifecycle domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.9 Rule Specification - ISPM-RULE-009

****Domain:**** governance

****Default Severity:**** medium

Intent

This rule detects a material identity posture condition in the governance domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.10 Rule Specification - ISPM-RULE-010

****Domain:**** monitoring

****Default Severity:**** high

Intent

This rule detects a material identity posture condition in the monitoring domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.11 Rule Specification - ISPM-RULE-011

****Domain:**** compliance

****Default Severity:**** critical

Intent

This rule detects a material identity posture condition in the compliance domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.12 Rule Specification - ISPM-RULE-012

****Domain:**** authentication

****Default Severity:**** low

Intent

This rule detects a material identity posture condition in the authentication domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.13 Rule Specification - ISPM-RULE-013

****Domain:**** authorization

****Default Severity:**** medium

Intent

This rule detects a material identity posture condition in the authorization domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.14 Rule Specification - ISPM-RULE-014

****Domain:**** lifecycle

****Default Severity:**** high

Intent

This rule detects a material identity posture condition in the lifecycle domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.

- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.15 Rule Specification - ISPM-RULE-015

****Domain:**** governance

****Default Severity:**** critical

Intent

This rule detects a material identity posture condition in the governance domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.16 Rule Specification - ISPM-RULE-016

****Domain:**** monitoring

****Default Severity:**** low

Intent

This rule detects a material identity posture condition in the monitoring domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.17 Rule Specification - ISPM-RULE-017

****Domain:**** compliance

****Default Severity:**** medium

Intent

This rule detects a material identity posture condition in the compliance domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.18 Rule Specification - ISPM-RULE-018

****Domain:**** authentication

****Default Severity:**** high

Intent

This rule detects a material identity posture condition in the authentication domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.19 Rule Specification - ISPM-RULE-019

Domain: authorization

Default Severity: critical

Intent

This rule detects a material identity posture condition in the authorization domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be

represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.20 Rule Specification - ISPM-RULE-020

****Domain:**** lifecycle

****Default Severity:**** low

Intent

This rule detects a material identity posture condition in the lifecycle domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.

- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.21 Rule Specification - ISPM-RULE-021

****Domain:**** governance

****Default Severity:**** medium

Intent

This rule detects a material identity posture condition in the governance domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.22 Rule Specification - ISPM-RULE-022

****Domain:**** monitoring

****Default Severity:**** high

Intent

This rule detects a material identity posture condition in the monitoring domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.23 Rule Specification - ISPM-RULE-023

****Domain:**** compliance

****Default Severity:**** critical

Intent

This rule detects a material identity posture condition in the compliance domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.24 Rule Specification - ISPM-RULE-024

Domain: authentication

Default Severity: low

Intent

This rule detects a material identity posture condition in the authentication domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.25 Rule Specification - ISPM-RULE-025

****Domain:**** authorization

****Default Severity:**** medium

Intent

This rule detects a material identity posture condition in the authorization domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.26 Rule Specification - ISPM-RULE-026

****Domain:**** lifecycle

****Default Severity:**** high

Intent

This rule detects a material identity posture condition in the lifecycle domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.27 Rule Specification - ISPM-RULE-027

****Domain:**** governance

****Default Severity:**** critical

Intent

This rule detects a material identity posture condition in the governance domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.

- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.28 Rule Specification - ISPM-RULE-028

****Domain:**** monitoring

****Default Severity:**** low

Intent

This rule detects a material identity posture condition in the monitoring domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.29 Rule Specification - ISPM-RULE-029

Domain: compliance

Default Severity: medium

Intent

This rule detects a material identity posture condition in the compliance domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.30 Rule Specification - ISPM-RULE-030

****Domain:**** authentication

****Default Severity:**** high

Intent

This rule detects a material identity posture condition in the authentication domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.31 Rule Specification - ISPM-RULE-031

****Domain:**** authorization

****Default Severity:**** critical

Intent

This rule detects a material identity posture condition in the authorization domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.32 Rule Specification - ISPM-RULE-032

Domain: lifecycle

Default Severity: low

Intent

This rule detects a material identity posture condition in the lifecycle domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be

represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.33 Rule Specification - ISPM-RULE-033

****Domain:**** governance

****Default Severity:**** medium

Intent

This rule detects a material identity posture condition in the governance domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.

- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.34 Rule Specification - ISPM-RULE-034

****Domain:**** monitoring

****Default Severity:**** high

Intent

This rule detects a material identity posture condition in the monitoring domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.35 Rule Specification - ISPM-RULE-035

****Domain:**** compliance

****Default Severity:**** critical

Intent

This rule detects a material identity posture condition in the compliance domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.36 Rule Specification - ISPM-RULE-036

****Domain:**** authentication

****Default Severity:**** low

Intent

This rule detects a material identity posture condition in the authentication domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.37 Rule Specification - ISPM-RULE-037

Domain: authorization

Default Severity: medium

Intent

This rule detects a material identity posture condition in the authorization domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.38 Rule Specification - ISPM-RULE-038

****Domain:**** lifecycle

****Default Severity:**** high

Intent

This rule detects a material identity posture condition in the lifecycle domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.39 Rule Specification - ISPM-RULE-039

****Domain:**** governance

****Default Severity:**** critical

Intent

This rule detects a material identity posture condition in the governance domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.40 Rule Specification - ISPM-RULE-040

****Domain:**** monitoring

****Default Severity:**** low

Intent

This rule detects a material identity posture condition in the monitoring domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.

- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.41 Rule Specification - ISPM-RULE-041

****Domain:**** compliance

****Default Severity:**** medium

Intent

This rule detects a material identity posture condition in the compliance domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.42 Rule Specification - ISPM-RULE-042

****Domain:**** authentication

****Default Severity:**** high

Intent

This rule detects a material identity posture condition in the authentication domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.43 Rule Specification - ISPM-RULE-043

****Domain:**** authorization

****Default Severity:**** critical

Intent

This rule detects a material identity posture condition in the authorization domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.44 Rule Specification - ISPM-RULE-044

****Domain:**** lifecycle

****Default Severity:**** low

Intent

This rule detects a material identity posture condition in the lifecycle domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.45 Rule Specification - ISPM-RULE-045

****Domain:**** governance

****Default Severity:**** medium

Intent

This rule detects a material identity posture condition in the governance domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be

represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.46 Rule Specification - ISPM-RULE-046

Domain: monitoring

Default Severity: high

Intent

This rule detects a material identity posture condition in the monitoring domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.

- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.47 Rule Specification - ISPM-RULE-047

****Domain:**** compliance

****Default Severity:**** critical

Intent

This rule detects a material identity posture condition in the compliance domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.48 Rule Specification - ISPM-RULE-048

****Domain:**** authentication

****Default Severity:**** low

Intent

This rule detects a material identity posture condition in the authentication domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.49 Rule Specification - ISPM-RULE-049

****Domain:**** authorization

****Default Severity:**** medium

Intent

This rule detects a material identity posture condition in the authorization domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.50 Rule Specification - ISPM-RULE-050

Domain: lifecycle

Default Severity: high

Intent

This rule detects a material identity posture condition in the lifecycle domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.51 Rule Specification - ISPM-RULE-051

****Domain:**** governance

****Default Severity:**** critical

Intent

This rule detects a material identity posture condition in the governance domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.52 Rule Specification - ISPM-RULE-052

****Domain:**** monitoring

****Default Severity:**** low

Intent

This rule detects a material identity posture condition in the monitoring domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.53 Rule Specification - ISPM-RULE-053

****Domain:**** compliance

****Default Severity:**** medium

Intent

This rule detects a material identity posture condition in the compliance domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.

- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.54 Rule Specification - ISPM-RULE-054

****Domain:**** authentication

****Default Severity:**** high

Intent

This rule detects a material identity posture condition in the authentication domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.55 Rule Specification - ISPM-RULE-055

****Domain:**** authorization

****Default Severity:**** critical

Intent

This rule detects a material identity posture condition in the authorization domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.56 Rule Specification - ISPM-RULE-056

****Domain:**** lifecycle

****Default Severity:**** low

Intent

This rule detects a material identity posture condition in the lifecycle domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.57 Rule Specification - ISPM-RULE-057

****Domain:**** governance

****Default Severity:**** medium

Intent

This rule detects a material identity posture condition in the governance domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.58 Rule Specification - ISPM-RULE-058

****Domain:**** monitoring

****Default Severity:**** high

Intent

This rule detects a material identity posture condition in the monitoring domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be

represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.59 Rule Specification - ISPM-RULE-059

Domain: compliance

Default Severity: critical

Intent

This rule detects a material identity posture condition in the compliance domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.

- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.

Appendix A.60 Rule Specification - ISPM-RULE-060

****Domain:**** authentication

****Default Severity:**** low

Intent

This rule detects a material identity posture condition in the authentication domain. It is included as an implementation template so the engineering team can construct versioned rule packs without ambiguity.

Inputs

- AQELYNIdentity canonical object.
- Provider entitlement and authentication metadata.
- Applicable policy controls.
- Baseline state for the identity scope.
- Evidence references created during discovery or assessment.

Evaluation Logic

The rule evaluator SHALL read only normalized fields, SHALL avoid provider-specific assumptions outside connector-owned transforms, and SHALL produce deterministic output for the same input set. Missing input shall be represented as unknown rather than inferred as compliant.

Finding Output

The output SHALL include finding_id, identity_id, rule_id, rule_version, severity, evidence_refs, explanation, recommended_action and compliance_refs. The explanation SHALL be safe for audit review and SHALL not expose secrets.

Test Cases

- Positive case: identity violates the rule and produces one finding.
- Negative case: identity is compliant and produces no finding.
- Unknown case: required evidence is missing and produces an informational evidence quality finding.
- Regression case: rule result remains stable across repeated evaluation.